

What Does Answer Change Rate Actually Measure? A Specificity Audit of Causal Intervention Tests for Explanation Faithfulness

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Abstract

A growing line of work tests whether large language model (LLM) explanations are *faithful* by intervening on the explanation (removing, perturbing, or swapping it) and checking whether the model’s final answer changes – an *Answer Change Rate* (ACR), read as causal evidence. We argue this inference requires a control existing work has not consistently reported: whether the answer changes comparably under an unrelated disturbance. We run the causal intervention paradigm on 12 models (6 matched base/instruction-tuned pairs, 8B–70B parameters) on GSM8K and MMLU-STEM, adding a content-free filler control matched for length, a re-injection baseline measuring re-prompting and decoding instability, and paired significance testing across conditions. McNemar’s test does not detect a significantly larger answer-change rate for *removal* than for the control in 14 of 24 model–task cells (58%) – an absence of detected difference, not equivalence – the re-prompting/decoding floor alone exceeds 80% answer-change in one model, and a commonly assumed instruction-tuning effect does not replicate across families. We do not conclude that LLM explanations are faithful or unfaithful; we conclude that ACR frequently fails a basic specificity check and cannot alone support causal-faithfulness claims. We release the full pipeline and propose a minimal reporting standard for future studies.

1 Introduction

Large language models (LLMs) often produce natural-language explanations alongside their predictions, and a substantial body of work asks whether these explanations are *faithful* – whether they causally reflect the process that produced the answer – or merely *plausible* post-hoc text (Turpin et al., 2023; Lyu et al., 2024). A natural test is

causal intervention: perturb the explanation while holding the question fixed, and measure whether the model’s final answer changes. If the answer rarely changes, the explanation is not doing causal work; if it often changes, the explanation is treated as influential.

This logic underlies a class of metrics built around an *Answer Change Rate* (ACR). Studies in this paradigm commonly report that ACR is higher on mathematical reasoning tasks (GSM8K) than on knowledge-intensive multiple-choice QA (MMLU-STEM), and some report that instruction-tuned models show lower ACR than their base counterparts. This argument has a basic gap: **a change in the answer following an intervention is only evidence about the explanation specifically if the model would *not* change its answer by a similar amount following an unrelated, content-free disturbance to the same prompt.** Without that comparison, ACR conflates two distinct properties: *sensitivity to explanation content* and *sensitivity to any disturbance of the prompt at this position*.

This paper takes that gap as its central research question. We construct an evaluation pipeline covering 12 models spanning 6 matched base/instruction-tuned pairs from three families (Qwen, Gemma, Llama) at 8B and 27–70B scale, with fixed evaluation sets of 500 examples per task, three perturbation seeds, and paired statistical testing. We add two controls absent from prior causal-intervention studies in this paradigm:

- **A content-free control** (control_irrelevant): the explanation slot is filled with fluent, topic-unrelated text of matched length, isolating sensitivity to *any* prompt disturbance from sensitivity to explanation content.

- **A re-injection baseline** (`reinject_original`): the model’s own unperturbed explanation is re-injected through the identical template. Because this still requires reconstructing the prompt and regenerating the answer, any observed answer change reflects a mix of decoding instability and re-prompting effects rather than decoding noise alone; it serves as a floor against which every other condition must be compared.

Our central finding is that McNemar’s test does not detect a significantly larger answer-change rate for removal than for the content-free control in **14 of 24 model–task cells (58%)** ($p > 0.05$); without a prespecified equivalence margin, this absence of a detected difference is not evidence that removal and the control produce equivalent answer-change behavior, and we do not interpret it as such. The re-injection floor ranges from $\approx 1\%$ to $\approx 83\%$ answer-change with *no* content-altering intervention at all, and a commonly assumed direction for the instruction-tuning effect reverses across model families under paired testing.

Contributions.

1. We identify a *specificity gap*: raw ACR does not distinguish sensitivity to explanation *content* from sensitivity to *any* disturbance at the explanation’s position.
2. We introduce two controls – content-free filler and a re-injection floor – and show both are necessary: the floor reaches 82.5% for one model, and McNemar’s test does not detect removal significantly exceeding the content-free control in 58% of cells.
3. We evaluate a commonly reported instruction-tuning effect with six matched pairs and paired testing, finding no consistent direction across families.
4. We release the full pipeline¹ so the specificity check can be applied routinely in future faithfulness studies.

2 Related Work

Explanation faithfulness is distinguished from plausibility in surveys categorizing evaluation methods by similarity, counterfactual, perturbation, and self-explanatory approaches (Lyu et al., 2024). CoT explanations are sensitive to biasing features not

reflected in stated reasoning (Turpin et al., 2023). Related work finds that correctness and faithfulness can diverge (Wallat et al., 2025), that explanation reliability depends on memory strength and evidence style (Li et al., 2025a), and that context-faithfulness can be improved with targeted alignment objectives (Bi et al., 2025). Prediction-Explanation consistency (Zhao and Daumé III, 2025) measures alignment between free-text explanations and model outputs. Concept-based perturbation separates faithfulness from readability (Li et al., 2024). Activation patching studies find alignment-tuned models can be more faithful (Yeo et al., 2025), while token-level self-explanation studies report substantial dataset-dependent variation (Fragkathoulas and Chlapanis, 2024). Further discussion of CoT effectiveness and faithfulness appears in Li et al. (2025b), and automatic NLG evaluators are shown to be inconsistent (Terentowicz et al., 2025).

A separate line of work studies contextual robustness: how model predictions respond to irrelevant or distracting content inserted into the prompt, without reference to explanation faithfulness. Shi et al. (2023) show LLM performance on grade-school math degrades substantially with irrelevant context added to the problem statement, introducing the GSM-IC benchmark. Wang and Zhao (2024) benchmark reasoning robustness across nine lexical, syntactic, and semantic perturbation types over fifteen reasoning datasets. Wang et al. (2025) use automated tree search to find model-adaptive distracting content, showing robustness estimates based on fixed perturbations can be overly optimistic. Our content-free control is a diagnostic borrowed from this tradition – a matched-surface-form disturbance analogous to a placebo condition – applied to test whether ACR-based faithfulness claims are confounded by the same generic sensitivity these robustness studies document.

3 Methodology

3.1 Models and Data

We evaluate 12 models spanning 6 matched base/instruction-tuned pairs across three families and two scale tiers: Qwen3-8B, Gemma-7B, and Llama-3.1-8B at the small (7–8B) tier, and Qwen2.5-32B, Gemma-2-27B, and Llama-3.1-70B at the large (27–70B) tier, each with one base and one instruction-tuned checkpoint (full model identifiers, context lengths, and licenses in Ta-

¹Code and run manifests: github.com/DURGESH716/Causal-Intervention-Tests

ble 2, Appendix C). All models use greedy decoding (temperature 0), global seed 42 for example sampling, and three independent perturbation seeds (1, 2, 3) for stochastic conditions. We evaluate on GSM8K (Cobbe et al., 2021) (multi-step mathematical reasoning) and 16 STEM subjects of MMLU (Hendrycks et al., 2021) (knowledge-intensive multiple-choice QA), both English-only. We fix $N = 500$ examples per task, write the exact sampled indices to disk before any model runs, and every model reads from this identical file. No training is performed; all models are used at inference time. The 70B Llama pair was evaluated under 4-bit quantization to fit available GPU memory; all other models run at full precision.

3.2 Explanation Conditions

Each example is first run under a chain-of-thought elicitation prompt to obtain the model’s original explanation and answer. Six conditions then regenerate the answer, each manipulating what occupies the `Explanation:` slot in the prompt while keeping all other text identical: **removal** – the `Explanation: ...` line is omitted entirely; all other instruction text is preserved. **perturb_word_shuffle** – words within the explanation are randomly shuffled at the word level (not sub-word tokens), preserving vocabulary and approximate length while destroying syntactic and logical structure. **perturb_step_reorder** – sentence-level steps are reordered while internal wording of each step is preserved, testing whether disruption of logical sequencing alone is sufficient to move the answer. **swap** – a different example’s explanation is substituted after a rule-based pass strips answer-revealing clauses, testing sensitivity to explanation identity rather than mere structural perturbation. **control_irrelevant** – fluent, topic-unrelated filler text of matched length replaces the explanation: same position, same surface statistics, no question-relevant content. This condition is the key specificity control: it tests whether any prompt disturbance at this position produces the same answer-change rate as the genuine perturbation conditions. **reinject_original** – the model’s own unperturbed explanation is re-injected through the identical template. Unlike the original elicitation, where the explanation and answer are generated jointly in a single decoding pass, this condition supplies the explanation as fixed prompt text and regenerates only the answer. Any resulting answer change therefore reflects a combination of decoding instability and

this change in conditioning (prompt reconstruction and re-parsing), not decoding noise in isolation; we treat it as a conservative floor against which all other conditions must be measured.

Filler construction. The content-free filler is built from a fixed bank of ten short, mundane, topic-unrelated sentences (e.g., library hours, bus routes, bakery recommendations) covering everyday civic and domestic subjects deliberately unrelated to mathematics or science, so that no filler instance can accidentally supply task-relevant content or an answer-like token. For each example, the bank is shuffled under the same per-example perturbation seed used by the other stochastic conditions, and sentences are concatenated in that shuffled order until the output reaches or exceeds the word count of the model’s own original explanation for that example, then truncated to match that count exactly; the filler therefore tracks each example’s explanation length individually rather than using a single fixed length across the dataset. We use one filler bank across all models and both tasks rather than per-model or per-task fillers, and we do not separately verify that the filler is free of stylistic or distributional mismatch relative to the explanations it replaces beyond length-matching and topic avoidance by construction; broader or adaptively-generated fillers, and an explicit check for answer-like tokens, are directions we did not pursue and flag as a limitation.

3.3 Prompt Templates

Base models receive raw string prompts with a 2–3-shot `Explanation: ... Answer: ...` header. Instruction-tuned models receive the same content wrapped through the tokenizer’s chat template. The elicitation prompt ends with `Explanation:` to elicit CoT before the answer. The intervention prompt takes the form `Question: {q} / Explanation: {e} / Answer:`, with the `Explanation:` line omitted entirely for removal. MMLU-STEM prompts include a `Choices: A/B/C/D` block. For Qwen3 instruct models, `enable_thinking=False` is set; any residual `<think>` block is stripped before answer extraction. We note that the elicitation prompt ends in `Explanation:` while every intervention prompt, including `reinject_original`, ends in `Answer:`; this shift in prompt boundary and conditioning is

shared by all intervention conditions and is one reason we treat `reinject_original` as a floor on answer-change rather than a zero-effect baseline.

3.4 Metrics

Raw ACR measures the fraction of examples on which the regenerated answer differs from the original answer, with 95% bootstrap confidence intervals (5,000 resamples):

$$\text{ACR}(\%) = \frac{\#(A_{\text{intervened}} \neq A_{\text{original}})}{N} \times 100 \quad (1)$$

where $A_{\text{intervened}}$ is the answer produced under the given intervention condition, A_{original} is the answer produced by the model under the unperturbed chain-of-thought prompt, and N is the total number of examples evaluated (500 per task). Examples for which the original answer cannot be parsed are excluded and their count is reported separately. For GSM8K, the first number appearing after the final “Answer:” marker is extracted, with fallback to the last number in the full output. For MMLU-STEM, a priority-ordered letter-extraction sequence is applied, followed by choice-text substring matching.

Seed aggregation. `removal` and `reinject_original` are deterministic (greedy decoding, single run) and require no aggregation. The four stochastic conditions (`perturb_word_shuffle`, `perturb_step_reorder`, `swap`, `control_irrelevant`) are each run under three independent perturbation seeds. Raw ACR (Equation 1) pools all three seed×example rows as independent observations before computing the change rate and its bootstrap CI. Noise-floor-adjusted ACR uses a different aggregation, appropriate to its paired design: for each example, the three seeds are first collapsed to a single change indicator by majority vote, and the paired bootstrap difference described above is then computed against the majority-voted `reinject_original` baseline. We report both rules explicitly because they are not interchangeable: pooling seeds as independent rows treats correlated repeated measurements of the same example as separate evidence, which is appropriate for a marginal rate estimate but not for the paired comparison the noise-floor adjustment requires.

Noise-floor-adjusted ACR is computed as the paired per-example difference between a con-

dition’s answer-change indicator and that of `reinject_original`, bootstrapped directly over examples. Negative values – indicating that a condition changes the answer *less often* than re-injecting the original explanation – are reported as-is rather than clipped to zero. Because `reinject_original` itself mixes decoding instability with re-prompting effects (see Section 3.3), this adjustment should be read as subtracting a conservative, possibly inflated floor, which if anything makes the adjusted values in Table 1 an *underestimate* of any residual explanation-specific effect.

Specificity test. We apply McNemar’s test (with continuity correction) between `removal` and `control_irrelevant` on paired examples. A non-significant result ($p > 0.05$) indicates that our test does not detect a difference in answer-change behavior between the two conditions at this sample size: we cannot conclude that removing the explanation produces more answer changes than inserting unrelated filler, and so no explanation-specific effect can be claimed for that cell. This is a failure to find evidence of a difference, not positive evidence that the two conditions have identical effects; we do not treat non-significance as proof of equivalence, and Table 1 reports the noise-floor-adjusted ACR for both conditions so that the size and direction of any observed gap can be inspected directly alongside the test outcome.

Effect size for the specificity test. McNemar’s test establishes whether REMOV and CTRL are distinguishable but not the size of any gap. We therefore additionally report, for GSM8K, a 95% bootstrap CI on the paired REMOV–CTRL difference for every model, computed with the same majority-vote-per-example and paired-bootstrap procedure used for the noise-floor adjustment (Table 3, Appendix). The CI excludes zero in 6 of 12 GSM8K cells; two of these (Llama-3.1-70B-Base and Qwen2.5-32B-Base) show a significant *negative* gap, meaning the content-free control changes the answer more than removal does at this sample size – consistent with, and giving a concrete magnitude to, the reversed-direction cells identified in Section 4.2.

Manual validation of answer extraction. We manually validated the answer-extraction heuristic on a random sample of 100 examples (50 per task). GSM8K extraction was correct in 48/50 cases (96.0%); MMLU-STEM extraction was cor-

rect in 40/50 cases (80.0%).

Transition breakdown. ACR is decomposed into right→wrong, wrong→right, and wrong→wrong transitions to distinguish interventions that disrupt correct reasoning from those that incidentally correct errors.

All inference is run with vLLM at greedy decoding and a maximum of 320 new tokens. A full run manifest (model IDs, seeds, timestamps) is released alongside the code.

4 Results

4.1 Noise-floor-adjusted ACR across all models and conditions

Table 1 reports noise-floor-adjusted ACR for all 12 models, both tasks, and all five intervention conditions, together with the raw re-injection floor (FLOOR) and the outcome of the McNemar specificity test (**Spec.**: * = significant at $p \leq 0.05$, i.e., removal produces significantly more answer changes than the content-free control; n.s. = non-significant, i.e., our test does not detect an explanation-specific effect at this sample size). Figures 1 and 2 show the same data as bar charts with 95% CIs, with small tier (left) and large tier (right) combined per task. Aggregated raw ACR is higher on GSM8K than MMLU-STEM for *removal* in 10 of 12 models, consistent with the intuition that explanation interventions move predictions more on multi-step arithmetic than knowledge recall. The qualitative task contrast survives the additional controls; the interpretation of *why* it survives does not, as Sections 4.2–4.4 show.

4.2 Removal is frequently not distinguishable from the control

In 14 of 24 model–task cells (58%), McNemar’s test does not detect a significant difference between removal and the content-free control at $p > 0.05$ (6/12 on GSM8K, 8/12 on MMLU-STEM), including both base and instruction-tuned models, both tasks, and both scale tiers. Failure to reject the null under McNemar’s test is not, by itself, evidence that removal and the control are practically equivalent: without a prespecified equivalence margin and a dedicated equivalence test (e.g., TOST), a non-significant result at this sample size is equally consistent with a genuine null effect and with an underpowered test missing a real, smaller difference. What we can say is narrower: in these cells, we cannot rule out that high raw ACR under

removal reflects generic prompt-position sensitivity rather than dependence on explanation content, and Table 1’s signed REMOV–CTRL gap together with Appendix D’s confidence intervals should be read alongside this test outcome to judge whether any such gap is also small in magnitude. We do not treat non-significance as proof that removal and the control have identical effects. Concretely, Gemma-2-27B-Base shows 74.8% raw ACR under removal and 69.6% under the content-free control ($p = 1.00$; corresponding noise-floor-adjusted values of 72.6 and 72.8 are given in Table 1); Llama-3.1-70B-Instruct shows 44.8% under removal and 45.7% under the content-free control ($p = 0.93$). In both cases the raw and adjusted values are close but not numerically identical, since the adjusted figure is a paired difference against each model’s own re-injection floor rather than the raw rate itself.

This failure is not uniform, and where the test *is* significant, the direction is not always the one a faithfulness reading would predict. Llama-3.1-8B-Base shows the clearest explanation-consistent effect on GSM8K ($p < 10^{-16}$; REMOV 87.4 vs. CTRL 68.2), while Qwen2.5-32B-Base is significant on GSM8K in the *opposite* direction ($p < 10^{-16}$; REMOV 11.0 vs. CTRL 60.0), reflecting the filler being more disruptive than removal. Reading the **Spec.** column together with the signed REMOV–CTRL gap in Table 1 is therefore necessary in every case: only cells where REMOV exceeds CTRL *and* the difference is significant support a removal-specific reading. By this stricter criterion, only 4 of 24 cells (Gemma-7B-Base and Llama-3.1-8B-Base on GSM8K, and Gemma-7B-Instruct on both tasks) show a significant, correctly-signed, removal-specific effect; the remaining 6 significant cells are significant in the *reversed* direction (the content-free control disturbs the answer more than removal), and the other 14 do not reach significance. The full per-cell breakdown, including which specific cells fall into each category, is given by the **Spec.** column and signed gap in Table 1 directly.

4.3 The re-injection floor is large and model-dependent

The FLOOR column of Table 1 shows `reinject_original` ACR ranging from 1.0% (Llama-3.1-70B-Base, GSM8K) to 82.5% (Gemma-7B-Instruct, GSM8K). As discussed in Section 3.3, this condition changes the prompt boundary and conditioning relative to the original elicitation, so it should be read as an upper-

Model	Task	Noise-floor-adjusted ACR (%)						Spec.
		REMOV	CTRL	WSHUF	SREORD	SWAP	FLOOR	
Gemma-7B-Base	GSM8K	84.0	58.9	40.3	3.0	93.6	6.0	*
Gemma-7B-Instruct	GSM8K	15.9	-18.1	6.9	-12.5	-9.5	82.5	*
Llama-3.1-8B-Base	GSM8K	87.4	68.2	37.0	1.4	71.2	1.6	*
Llama-3.1-8B-Instruct	GSM8K	5.6	7.2	10.0	7.0	9.0	45.2	n.s.
Qwen3-8B-Base	GSM8K	17.6	53.3	20.2	2.6	96.6	3.4	*
Qwen3-8B-Instruct	GSM8K	7.9	8.3	9.7	2.0	9.3	1.8	n.s.
Gemma-2-27B-Base	GSM8K	72.6	72.8	26.4	0.8	91.0	2.2	n.s.
Gemma-2-27B-Instruct	GSM8K	4.4	6.2	4.2	3.2	5.6	4.0	n.s.
Llama-3.1-70B-Base	GSM8K	67.2	73.6	23.8	2.4	96.8	1.0	*
Llama-3.1-70B-Instruct	GSM8K	15.6	16.0	18.8	6.2	15.8	29.2	n.s.
Qwen2.5-32B-Base	GSM8K	11.0	60.0	14.2	0.2	76.2	1.2	*
Qwen2.5-32B-Instruct	GSM8K	29.0	28.2	28.0	19.8	23.6	24.6	n.s.
Gemma-7B-Base	MMLU-STEM	25.1	34.6	14.2	2.8	36.4	14.2	*
Gemma-7B-Instruct	MMLU-STEM	6.6	-3.9	2.9	-1.0	2.9	46.2	*
Llama-3.1-8B-Base	MMLU-STEM	30.7	32.9	16.3	3.3	36.6	11.4	n.s.
Llama-3.1-8B-Instruct	MMLU-STEM	23.7	25.2	20.7	7.8	31.1	9.1	n.s.
Qwen3-8B-Base	MMLU-STEM	19.9	21.3	7.6	0.8	31.6	4.8	n.s.
Qwen3-8B-Instruct	MMLU-STEM	19.5	24.0	23.8	10.2	43.3	18.5	*
Gemma-2-27B-Base	MMLU-STEM	15.2	16.4	5.0	1.2	20.8	11.2	n.s.
Gemma-2-27B-Instruct	MMLU-STEM	-1.4	0.5	-0.5	3.1	-1.6	18.8	n.s.
Llama-3.1-70B-Base	MMLU-STEM	20.0	21.2	9.5	-0.6	24.8	9.1	n.s.
Llama-3.1-70B-Instruct	MMLU-STEM	17.6	19.2	17.6	4.1	19.4	6.7	n.s.
Qwen2.5-32B-Base	MMLU-STEM	16.8	26.5	6.5	0.6	33.3	2.2	*
Qwen2.5-32B-Instruct	MMLU-STEM	20.8	25.8	16.2	2.5	54.6	11.2	n.s.

Table 1: Noise-floor-adjusted ACR (%) for all 12 models on GSM8K (top) and MMLU-STEM (bottom), relative to the `reinject_original` floor (negative values change the answer *less* than re-injection). Columns: REMOV = removal; CTRL = content-free filler; WSHUF = word shuffle; SREORD = step reorder; SWAP = cross-example substitution; FLOOR = raw re-injection ACR. **Spec.:** McNemar’s test between REMOV and CTRL (*: $p \leq 0.05$; n.s.: $p > 0.05$); significance does not indicate direction (e.g. Qwen2.5-32B-Base/GSM8K has CTRL > REMOV despite * – see Section 4.2).

bound floor combining decoding instability and re-prompting effects rather than decoding noise in isolation. Even under this conservative reading, an 82.5% floor means that most of what would naively be reported as “the explanation causally drives over 80% of predictions” is, in fact, indistinguishable from how often this model changes its answer when its own explanation is simply re-fed through the same pipeline. Instruction-tuned models show higher floors than their base counterparts in all six matched pairs (per-pair average across tasks), suggesting that the fluency and confidence of aligned generation does not correspond to more stable answer commitments – itself a finding of direct relevance to faithfulness research, since deployment decisions are often made on the basis of such surface-level fluency.

4.4 The instruction-tuning effect has no consistent direction

A claim common in this literature is that instruction tuning reduces causal dependence on explanations, based on single base/instruct comparisons.

With six matched pairs and paired McNemar testing across all six conditions and both tasks (72 pair-task-condition cells), 58 cells reach significance ($p \leq 0.05$); of these, the instruction-tuned model shows *higher* ACR in 32 (55%) and *lower* ACR in 26 (45%) – close to a coin flip. Restricting to `removal` specifically, instruction-tuned models show lower ACR in 7 of 12 pairs. Individual pairs diverge strongly in opposite directions: Gemma-2-27B shows base 74.8% vs. instruct 8.4% on GSM8K removal ($p < 10^{-70}$), while Qwen2.5-32B reverses just as strongly (base 12.2% vs. instruct 53.6%, $p < 10^{-35}$). There is no family- or scale-independent instruction-tuning effect on ACR in either direction.

4.5 Swap ACR and transition breakdown

Despite answer-revealing clauses being stripped from donor explanations, `swap` remains the highest-ACR condition for base models on GSM8K, exceeding 90% for three models (Gemma-7B-Base 98.4%, Llama-3.1-70B-Base 96.5%, Qwen3-8B-Base 98.7%). As the leakage-stripping

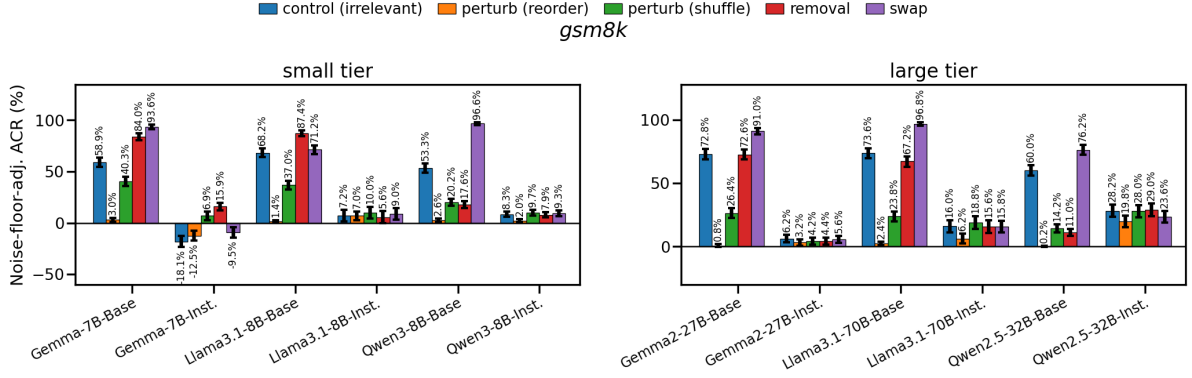


Figure 1: Noise-floor-adjusted ACR on GSM8K, small tier 7–8B (left) and large tier 27–70B (right). The `control_irrelevant` bar is comparable in magnitude to `removal` for several base models in both tiers; `perturb_step_reorder` is near-zero throughout.

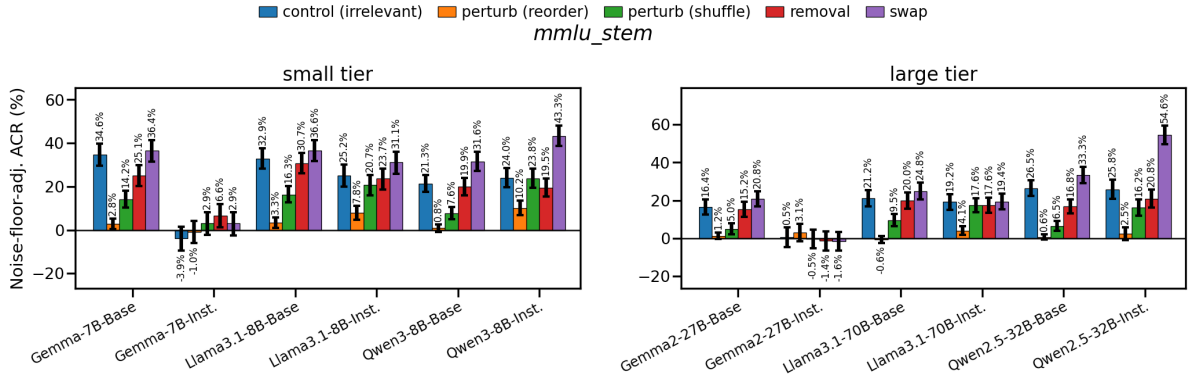


Figure 2: Noise-floor-adjusted ACR on MMLU-STEM, small tier (left) and large tier (right). ACR values are more moderate than GSM8K across all conditions, and the gap between `removal` and `control_irrelevant` is consistently smaller.

heuristic is regex-based and best-effort, residual leakage cannot be fully ruled out.

The transition breakdown under `removal` further clarifies what ACR represents. For high-ACR base models on GSM8K, the dominant transition is right→wrong: Gemma-2-27B-Base 49.6%, Llama-3.1-70B-Base 48.7%, and Llama-3.1-8B-Base 45.6%, confirming that removal disrupts *previously correct* reasoning rather than merely reshuffling errors. For Gemma-7B-Instruct – with its 82.5% re-injection floor – the majority of changes under removal are wrong→wrong (80.8%), consistent with decoding instability and re-prompting effects rather than genuine explanation dependence. On MMLU-STEM, the breakdown is more mixed: right→wrong and wrong→right transitions are comparable in magnitude (e.g. Llama-3.1-8B-Base: 15.7% vs. 14.4%), indicating that removal both disrupts and accidentally corrects reasoning in roughly equal measure.

Why SREORD is near-zero. Figures 1 and 2 show `perturb_step_reorder` near zero across almost every model and task. We quantified whether this reflects a weak perturbation or a perturbation with little to act on: for each `perturb_step_reorder` record we compared the sentence order of the reordered explanation actually inserted into the prompt against the original explanation’s sentence order. Pooled across all 24 model–task cells, 15.7% of explanations had fewer than two sentences and were therefore unchanged by construction (a genuine no-op, ranging from 0% to 50.8% by cell); among explanations with two or more sentences, the sentence order differed from the original in 99.9% of cases, since a uniform random permutation of more than a few sentences has a vanishingly small chance of coinciding with the original order. SREORD’s near-zero ACR is therefore not explained by the perturbation failing to alter the explanation – it reli-

ably does, wherever there is more than one sentence to reorder – but by sentence-level reordering alone being insufficient to move the model’s final answer, in contrast to `perturb_word_shuffle` and `swap`, which alter local wording or content and produce substantially higher ACR in the same cells.

5 Discussion

The pattern across Sections 4.2–4.4 is consistent: every claim we subjected to a specificity check or paired significance test came back partially or fully unsupported, while the basic task contrast (GSM8K > MMLU-STEM) survived. We read this as evidence about the *metric*: ACR without a content-free control and a re-injection floor is not sufficiently specific to support a causal-faithfulness inference. A model can show high ACR under removal because it is disturbed by an unusual gap in the prompt, or because its answer commitment is unstable regardless of prompt content – both produce the same observable signature the literature has read as faithfulness. Raw ACR must therefore be reported alongside, at minimum, the two controls introduced here, with conclusions scoped to cells where the specificity test passes *and* the direction of the effect is explanation-consistent (Section 4.2); we did not find a single headline claim that holds uniformly once these controls are applied.

Why does removal fail the specificity test so often? We identify three plausible, confounded mechanisms – positional sensitivity, answer commitment instability, and explanation non-use – separable in principle but not disentangled here (Appendix B); we leave mechanistic disentanglement to future work.

The task contrast is real but differently explained. The finding that ACR is higher on GSM8K than MMLU-STEM survives all controls, but admits an alternative to the usual explanation-necessity reading: GSM8K’s open numeric answer space is inherently more sensitive to prompt perturbations than MMLU-STEM’s four fixed choices, which constrain generation under distribution shifts. This connects to a broader contextual-robustness literature finding that irrelevant content shifts model outputs independent of any faithfulness-relevant mechanism (Shi et al., 2023; Wang and Zhao, 2024; Wang et al., 2025); our task contrast may partly

reflect this generic sensitivity rather than a true difference in explanation dependence.

Practical and deployment implications. Instruction-tuned models show higher re-injection floors than base models across all six pairs, suggesting alignment improves surface fluency without stabilizing answer commitment – floor measurement should be a first-class diagnostic, not normalized away. “Faithful”/“unfaithful” claims should not alone justify deployment decisions in education, scientific QA, or decision support: high raw ACR may in fact reflect less explanation-dependence than a competitor’s lower raw ACR. Our controls cost one extra inference pass, making them practical to add to existing pipelines. Appendix B discusses mechanistic interpretability connections and `swap` condition limitations.

6 Minimal Reporting Standard for ACR-Based Faithfulness Studies

We articulate a minimal reporting standard of five conditions that we argue are *necessary* (though not sufficient) before raw ACR results can be interpreted as evidence of explanation faithfulness, each addressing a specific threat to validity identified in this paper: (1) report a content-free specificity control paired with a significance test against the primary intervention; (2) report and adjust for a re-injection floor rather than treating raw ACR as decoding-noise-free; (3) use paired significance testing throughout, and do not treat non-significance as evidence of equivalence without a dedicated equivalence test; (4) report the answer-transition breakdown (right→wrong, wrong→right, wrong→wrong) rather than aggregate ACR alone; and (5) pre-register sampled indices and release full run manifests. Full descriptions of each condition are given in Appendix A.

7 Conclusion

We studied the causal-intervention paradigm for LLM explanation faithfulness across 12 models using fixed sampling, paired testing, and two novel controls: a content-free disturbance control and a re-injection floor. The standard ACR metric fails a basic specificity check – or shows the wrong direction – in most model–task cells, and the commonly assumed instruction-tuning effect does not replicate beyond a single family. We release our full pipeline to enable routine specificity checks.

Limitations

Our headline figures are empirical, not universal constants, and may vary across domains, languages, or rationale formats. The 58% non-significance rate (Section 4.2) reflects a failure to detect a significant, explanation-consistent effect at $N = 500$ per cell, not evidence of equivalence; we apply no formal equivalence test (e.g., TOST) with a prespecified margin, so no claim here should be read as establishing statistical equivalence between removal and the content-free control. Both evaluation sets are English-only; the `swap` condition’s leakage mitigation is heuristic. Our single content-free filler bank (Section 3.2) is not independently verified against stylistic or distributional mismatch beyond length-matching and topic avoidance by construction; alternative or adaptively-generated fillers remain future work. `reinject_original` mixes decoding instability with re-prompting effects (Section 3.3): we show *whether* ACR is explanation-specific, not *what* drives instability where the test fails. 4-bit quantization on the 70B pair may introduce precision artifacts we cannot rule out. We also do not systematically analyze how specificity failures or floor magnitudes vary with model scale beyond the two tiers evaluated, nor do we extend beyond GSM8K and MMLU-STEM to other reasoning styles (e.g., commonsense or planning); both are natural directions for future work given the model and task coverage already collected here.

Ethical considerations

This work re-evaluates public models on public benchmarks and introduces no new human-subject data, model training, or sensitive data; all models are used per their published licenses (Table 2). 4-bit quantization was a hardware necessity. The net effect is to reduce over-trusting of LLM explanations based on an under-specified metric.

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A Minimal Reporting Standard: Full Conditions

Condition 1: Report a content-free specificity control. Every ACR study should include a content-free condition (explanation slot filled with fluent, topic-unrelated text of matched length) and a paired significance test (e.g., McNemar’s) against the primary intervention. A non-significant result

means the study cannot claim sensitivity to explanation content specifically – only that it has not detected sensitivity beyond *something* at that prompt position. We recommend also reporting the signed difference between the two conditions (Table 1), since a non-significant test with a small or reversed gap supports a different conclusion than a non-significant test with a large gap in the explanation-consistent direction. This costs one extra inference pass per example.

Condition 2: Report and adjust for a re-injection floor. Studies should measure `reinject_original` ACR – the answer-change rate when the model’s own explanation is fed back through the same regeneration pipeline – and report it alongside every headline figure, together with a description of how re-injection differs from the original elicitation (e.g., prompt boundary, conditioning; Section 3.3). Models with a floor above 20% should report floor-adjusted ACR (paired difference from the floor) rather than raw ACR, since a substantial share of the observed change rate may reflect decoding instability or re-prompting effects rather than the intervention itself.

Condition 3: Use paired significance testing throughout. Comparisons between conditions and between models (e.g., base vs. instruct) should use paired tests on example-level outcomes, not point estimates alone. McNemar’s test suits paired binary outcomes; bootstrap CIs add a complementary uncertainty view. Non-significance should be reported as such – an absence of detected difference – rather than as evidence of equivalence; where a claim of practical equivalence is intended, a pre-specified equivalence margin and an equivalence test (e.g., TOST) should be used instead of relying on a non-significant difference test. This matters most for instruction-tuning comparisons, where a single model-pair can suggest the opposite conclusion from a multi-pair analysis.

Condition 4: Report the answer-transition breakdown. Aggregate ACR should be decomposed into right→wrong, wrong→right, and wrong→wrong transitions. A 40% ACR composed mostly of wrong→wrong transitions (unstable decoding between incorrect answers) implies a different conclusion than one composed mostly of right→wrong transitions (genuine disruption of correct reasoning).

Condition 5: Pre-register indices and release run manifests. Sampled example indices should be written to disk before evaluation and released with results to prevent post-hoc subsetting. Full run manifests (model IDs, software versions, seeds, timestamps) should accompany released code to support exact replication.

B Extended Discussion

Why does removal fail the specificity test so often? We identify three plausible mechanisms, which future work should disentangle mechanistically. First, *positional sensitivity*: the explanation slot occupies a consistent position in the prompt, and its absence or substitution may activate distribution-shift sensitivity independent of content. Second, *answer commitment instability*: for models with high re-injection floors (Table 1, FLOOR column), no stable answer is maintained regardless of the explanation provided. These two are confounded in raw ACR but separable using our controls: content-free filler isolates positional sensitivity; re-injection isolates commitment instability, modulo the re-prompting effects discussed in Section 3.3. Third, *explanation non-use*: some models may treat the explanation as a weak prior rather than a causal input. All three predict high raw ACR under removal; only the first two predict it under content-free filler.

Connection to mechanistic interpretability. Our behavioral paradigm complements mechanistic findings from activation patching (Yeo et al., 2025) and token-level self-explanation (Fragkathoulas and Chlapanis, 2024), which examine internal representations. Model-task cells passing our specificity test in the explanation-consistent direction (e.g., Llama-3.1-8B-Base on GSM8K, Gemma-7B-Instruct on both tasks) are promising targets for such mechanistic follow-up.

Limitations of the swap condition. The `swap` condition evaluates sensitivity to explanation identity by using a held-out explanation from an unrelated example. However, high base model ACR on GSM8K (> 90% for three models) remains ambiguous: it could indicate genuine sensitivity to mismatched content or residual leakage from our heuristic, unvalidated regex-based answer stripping. Consequently, `swap` results require cautious interpretation, and we place less interpretive weight on this condition than on `removal`

or `control_irrelevant`.

C Full Model List

D Effect Sizes for the Specificity Test

Model	HF Repo	Params	Type	Context	License
Qwen3-8B-Base	Qwen/Qwen3-8B-Base	8B	Base	32k	Apache-2.0
Qwen3-8B-Instruct	Qwen/Qwen3-8B	8B	Instruct	32k	Apache-2.0
Gemma-7B-Base	google/gemma-7b	7B	Base	8k	Gemma
Gemma-7B-Instruct	google/gemma-7b-it	7B	Instruct	8k	Gemma
Llama-3.1-8B-Base	meta-llama/Llama-3.1-8B	8B	Base	128k	Llama 3.1
Llama-3.1-8B-Instruct	meta-llama/Llama-3.1-8B-Instruct	8B	Instruct	128k	Llama 3.1
Qwen2.5-32B-Base	Qwen/Qwen2.5-32B	32B	Base	128k	Apache-2.0
Qwen2.5-32B-Instruct	Qwen/Qwen2.5-32B-Instruct	32B	Instruct	128k	Apache-2.0
Gemma-2-27B-Base	google/gemma-2-27b	27B	Base	8k	Gemma
Gemma-2-27B-Instruct	google/gemma-2-27b-it	27B	Instruct	8k	Gemma
Llama-3.1-70B-Base	meta-llama/Llama-3.1-70B	70B	Base	128k	Llama 3.1
Llama-3.1-70B-Instruct	meta-llama/Llama-3.1-70B-Instruct	70B	Instruct	128k	Llama 3.1

Table 2: All 12 evaluated model checkpoints, organized by family (Qwen, Gemma, Llama) and scale tier (small: 7–8B; large: 27–70B), with one base and one instruction-tuned model per pair. The 70B Llama pair was evaluated under 4-bit quantization to fit available GPU memory; all other models run at full precision. HF Repo gives the Hugging Face identifier used to load each checkpoint.

Model	REMOV-CTRL	95% CI	<i>N</i>
Gemma-2-27B-Base	−0.2	[−4.0, +3.4]	500
Gemma-2-27B-Instruct	−1.8	[−4.0, +0.4]	498
Gemma-7B-Base	+25.1	[+20.6, +29.7]	499
Gemma-7B-Instruct	+34.1	[+29.5, +38.4]	464
Llama-3.1-70B-Base	−6.4	[−10.2, −2.6]	500
Llama-3.1-70B-Instruct	−0.4	[−4.6, +3.8]	500
Llama-3.1-8B-Base	+19.2	[+15.2, +23.4]	500
Llama-3.1-8B-Instruct	−1.6	[−5.8, +2.8]	500
Qwen2.5-32B-Base	−49.0	[−53.6, −44.4]	500
Qwen2.5-32B-Instruct	+0.8	[−4.0, +5.8]	500
Qwen3-8B-Base	−35.7	[−40.9, −30.7]	499
Qwen3-8B-Instruct	−0.4	[−2.4, +1.6]	494

Table 3: 95% bootstrap CI on the paired REMOV-CTRL difference (percentage points) for GSM8K, per model, computed with the same majority-vote-per-example and paired-bootstrap procedure as the noise-floor adjustment (Section 3.4). *N* is the number of examples with a valid original answer used in the paired comparison. 6 of 12 CIs exclude zero.